

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL A-LEVEL CHEMISTRY (9620)

Unit 3: Inorganic 2 and Physical 2

Wednesday 5 June 2024 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- the Periodic Table/Data Sheet, provided as an insert
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do **not** write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- All working must be shown.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 80.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
TOTAL	



J U N 2 4 C H 0 3 0 1

Answer **all** questions in the spaces provided.

0 1

This question is about the enthalpy of lattice dissociation of calcium bromide, CaBr_2

0 1 . 1

Define enthalpy of lattice dissociation.

[2 marks]

Table 1 shows some enthalpy change data.

Table 1

	$\Delta H / \text{kJ mol}^{-1}$
Enthalpy of solution of CaBr_2	–110
Enthalpy of hydration of Ca^{2+}	–1650
Enthalpy of hydration of Br^-	–335

0 1 . 2

Use the data in **Table 1** to calculate the enthalpy of lattice dissociation, in kJ mol^{-1} , for calcium bromide.

[2 marks]

Enthalpy of lattice dissociation _____ kJ mol^{-1}

0 1 . 3

The enthalpy of lattice dissociation for calcium bromide calculated in Question **01.2** is the experimental value.

Explain why this experimental value is different from the theoretical value for the enthalpy of lattice dissociation of calcium bromide.

[1 mark]



0 1 . 4

A student does an experiment to determine the enthalpy of solution for calcium bromide.

1.25 g of solid calcium bromide is added from a weighing boat to 25.0 cm³ distilled water in a plastic (polystyrene) cup and the mixture is stirred gently.

When the calcium bromide dissolves, the temperature of the mixture rises from 19.6 °C to 24.9 °C

Calculate the enthalpy of solution, in kJ mol⁻¹, for calcium bromide.

Assume that the specific heat capacity of the solution is 4.18 J K⁻¹ g⁻¹

Assume that the density of the solution is 1.00 g cm⁻³

[3 marks]

Enthalpy of solution _____ kJ mol⁻¹

0 1 . 5

Another student repeats the experiment.

After the calcium bromide is added into the plastic cup, the student washes the weighing boat with a small volume of distilled water.

The student adds the washings to the plastic cup to ensure that all the calcium bromide is transferred.

Explain why adding the washings causes inaccurate results.

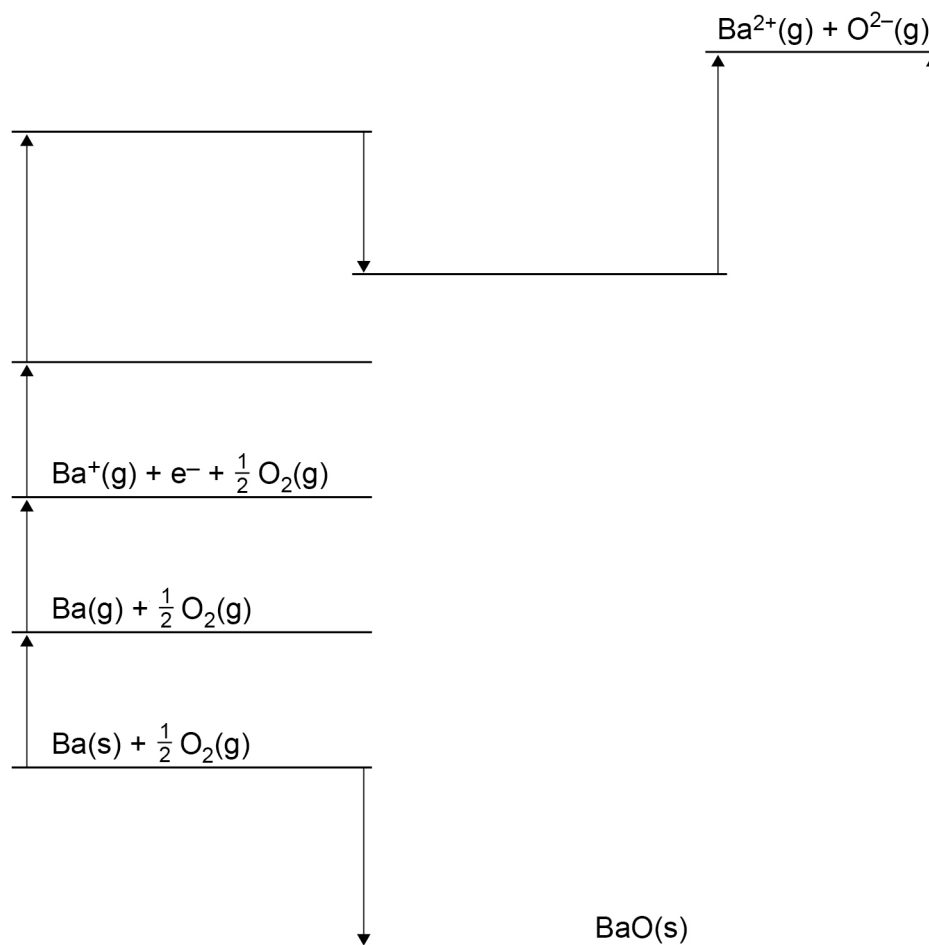
[1 mark]

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0 2

Figure 1 shows an incomplete Born–Haber cycle for barium oxide, BaO

Figure 1



0 2 . 1

Write formulas, including state symbols, of the appropriate species on each of the three blank horizontal lines in **Figure 1**.

[3 marks]



Table 2 shows some enthalpy change data.

Table 2

	$\Delta H / \text{kJ mol}^{-1}$
Enthalpy of atomisation of barium	+176
Enthalpy of atomisation of oxygen	+248
First ionisation enthalpy of barium	+502
Second ionisation enthalpy of barium	+966
First electron affinity of oxygen	-142
Second electron affinity of oxygen	+844
Lattice dissociation enthalpy of barium oxide	+3152

0 2 . 2

Use **Figure 1** and **Table 2** to calculate the enthalpy of formation, in kJ mol^{-1} , of barium oxide.

[2 marks]

Enthalpy of formation _____ kJ mol^{-1}

5

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0 3

Magnesium reacts with carbon dioxide.

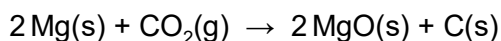


Table 3 shows how the value for the Gibbs free-energy change (ΔG) for this reaction changes with temperature.

Table 3

Temperature / K	$\Delta G / \text{kJ mol}^{-1}$
200	-165
400	-122
500	-101
700	-58
900	-15

0 3 . 1

Use the data in **Table 3** to plot a graph on **Figure 2** on **page 7** of ΔG (y-axis) against temperature.

Draw a line of best fit.

[2 marks]

0 3 . 2

Use your graph on **Figure 2** on **page 7** to deduce the enthalpy change, in kJ mol^{-1} , for the reaction.

[1 mark]

Enthalpy change _____ kJ mol^{-1}

0 3 . 3

Use your graph on **Figure 2** on **page 7** to determine the entropy change, in $\text{J K}^{-1} \text{mol}^{-1}$, for the reaction.

[2 marks]

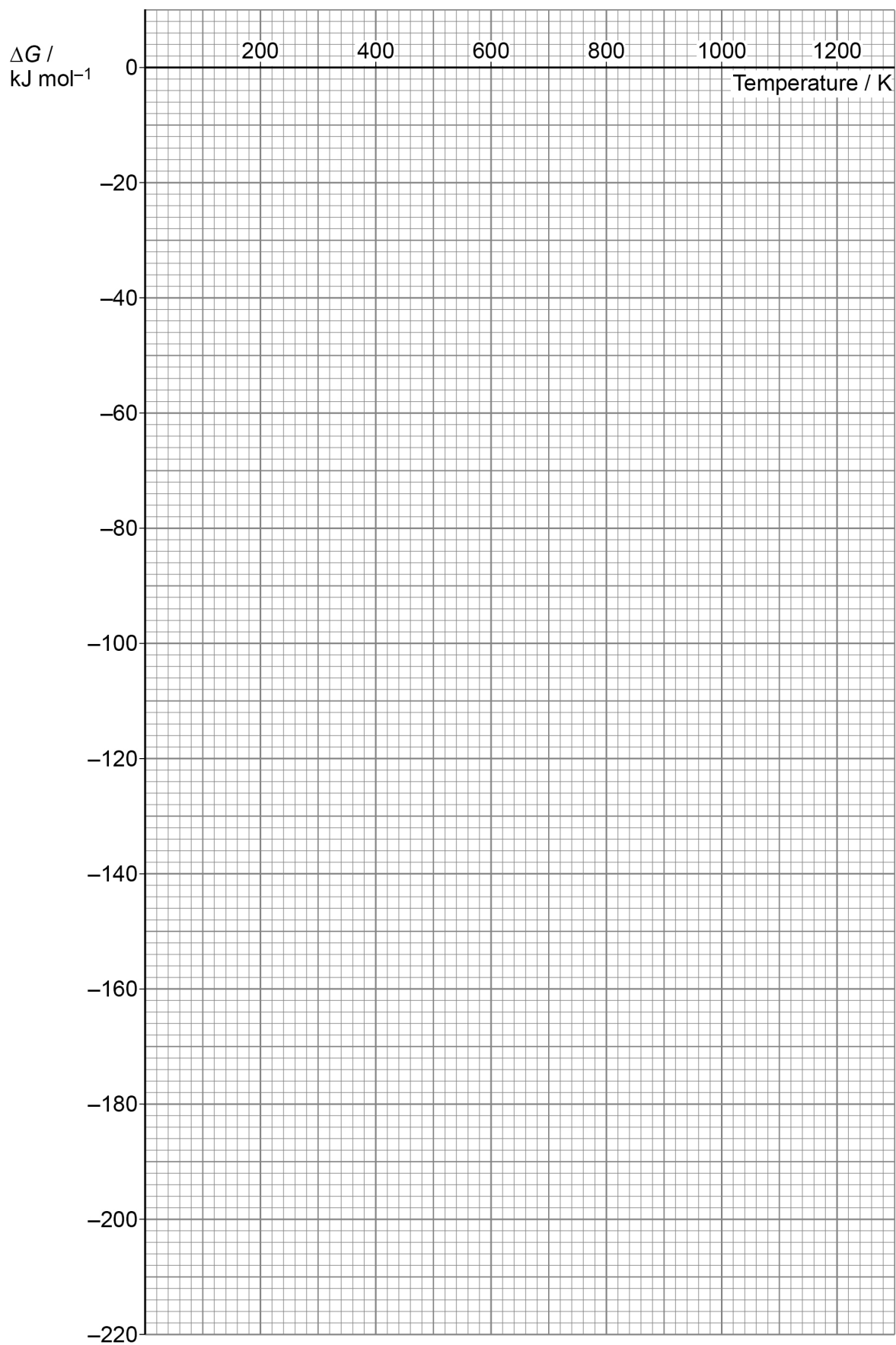
Entropy change _____ $\text{J K}^{-1} \text{mol}^{-1}$

0 3 . 4

The gradient of the line on **Figure 2** changes at 923 K. Suggest why.

[1 mark]



Figure 2

Use your completed **Figure 2** to answer Questions **03.2** and **03.3** on **page 6**.

Turn over ►

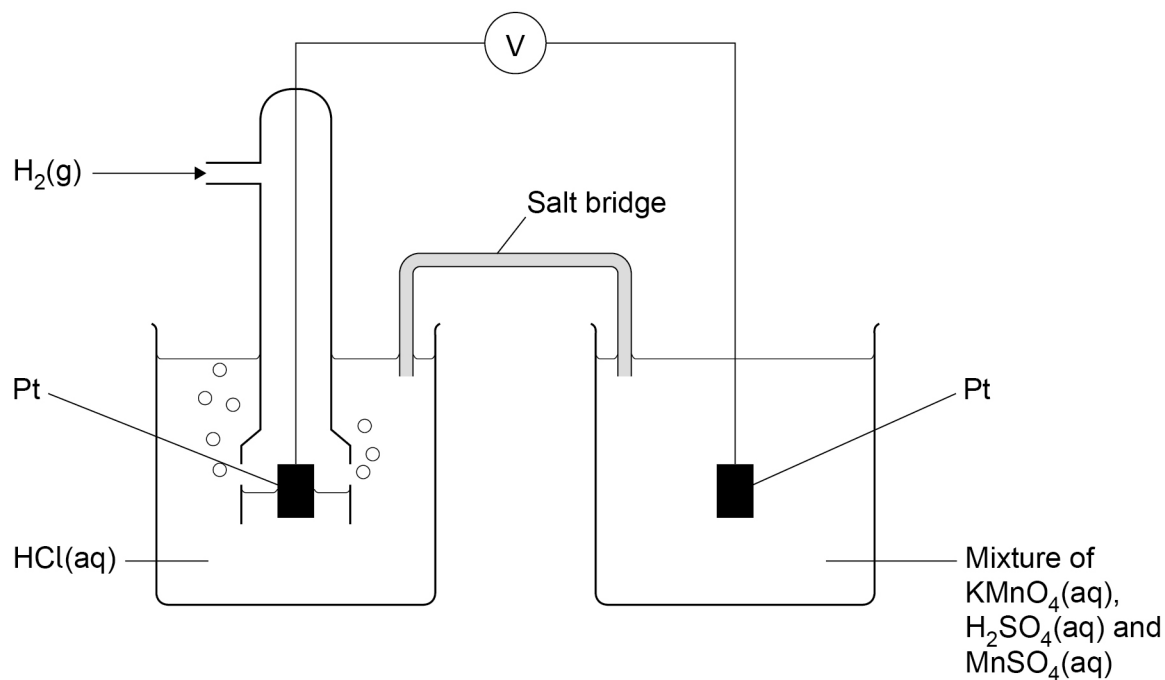


0 4

Figure 3 shows the electrochemical cell used to measure the standard electrode potential for the $\text{MnO}_4^- / \text{Mn}^{2+}$ electrode.

In this electrochemical cell, the standard hydrogen electrode is shown on the left-hand side.

Figure 3



0 4 . 1

State **three** conditions needed for the standard hydrogen electrode.

[2 marks]

0 4 . 2

Give the conventional cell representation for this electrochemical cell.

[2 marks]



Table 4 shows some electrode potential data.

Table 4

	$E^\ominus / \text{V}$
$\text{MnO}_4^-(\text{aq}) + 8 \text{H}^+(\text{aq}) + 5 \text{e}^- \rightarrow \text{Mn}^{2+}(\text{aq}) + 4 \text{H}_2\text{O}(\text{l})$	+1.52
$\text{Cl}_2(\text{g}) + 2 \text{e}^- \rightarrow 2 \text{Cl}^-(\text{aq})$	+1.36
$2 \text{H}^+(\text{aq}) + 2 \text{e}^- \rightarrow \text{H}_2(\text{g})$	0.00
$\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow \text{Na}(\text{s})$	-2.71

0 4 . 3

Use data from **Table 4** to explain why sodium chloride solution should **not** be used in the salt bridge in **Figure 3**.

[2 marks]

0 4 . 4

The standard hydrogen electrode is replaced by a $\text{Fe}^{3+} / \text{Fe}^{2+}$ electrode.

The reading on the voltmeter is 0.75 V

Calculate the standard electrode potential of the $\text{Fe}^{3+} / \text{Fe}^{2+}$ electrode.

[1 mark]

$E^\ominus$ _____ V

Question 4 continues on the next page

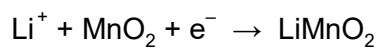
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Lithium is used to make rechargeable batteries.

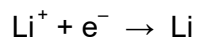
The half-equations at each of the electrodes in one type of lithium cell are:

Positive Electrode



$$E^\ominus = -0.15 \text{ V}$$

Negative Electrode



$$E^\ominus = -3.05 \text{ V}$$

0 4 . 5 Deduce the overall equation for the reaction when this cell is being **recharged**.

[1 mark]

0 4 . 6 Suggest why there must be no water in this cell.

[1 mark]

9



0 5

This question is about acids and bases.

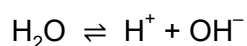
0 5 . 1

Calculate the pH of a $0.150 \text{ mol dm}^{-3}$ solution of hydrochloric acid.
Give your answer to 2 decimal places.

[1 mark]

pH _____

Water dissociates as shown

**0 5 . 2**Give an expression for K_w **[1 mark]** K_w **0 5 . 3**

Calculate the pH of a $0.250 \text{ mol dm}^{-3}$ aqueous solution of sodium hydroxide at 30°C

The value of K_w is $1.47 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$ at 30°C

Give your answer to 2 decimal places.

[2 marks]

pH _____

0 5 . 4

When the aqueous solution of sodium hydroxide in Question **05.3** is heated, the pH of the solution decreases.

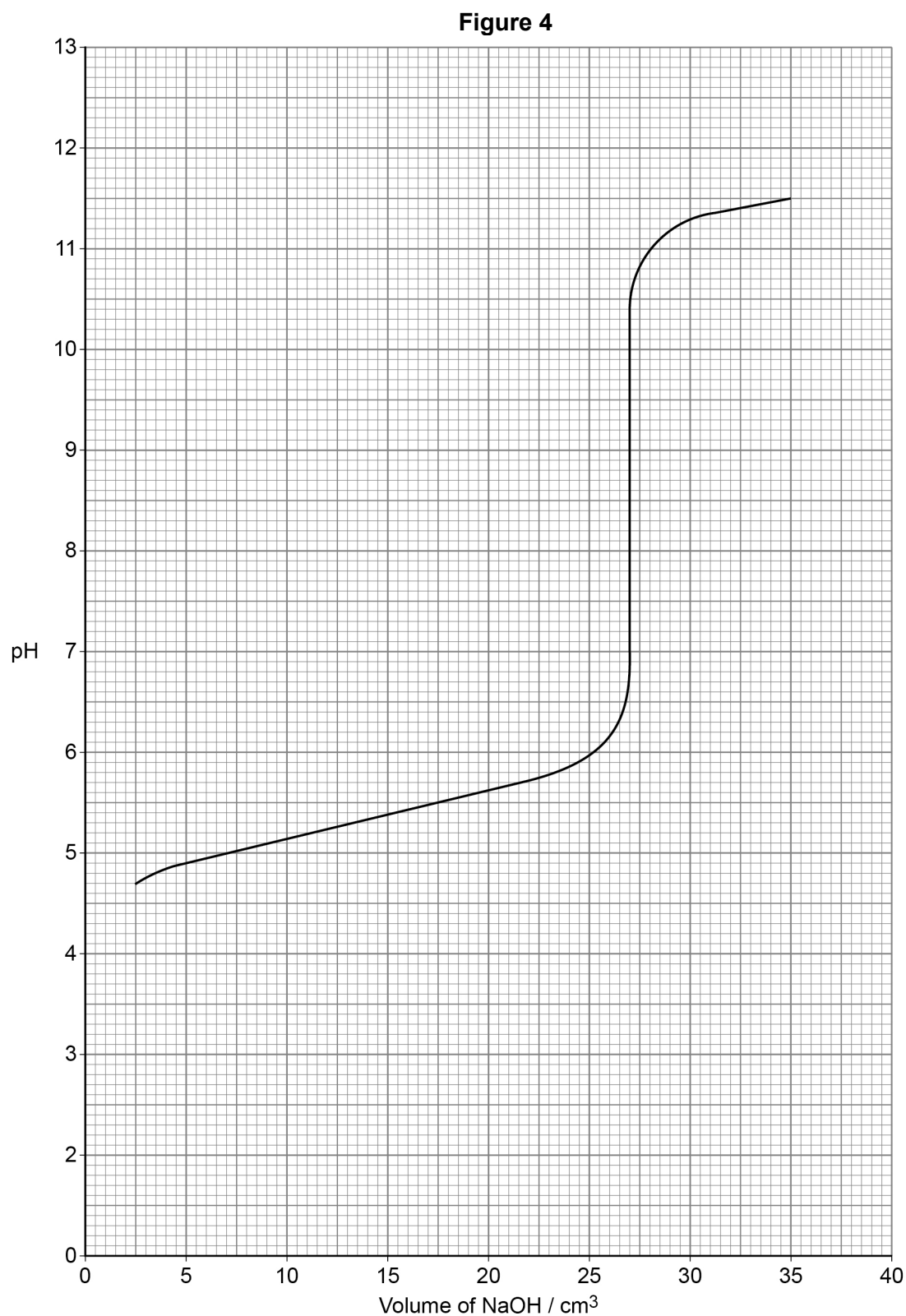
Explain why.

[2 marks]

Turn over ►

A student adds $0.100 \text{ mol dm}^{-3}$ aqueous sodium hydroxide from a burette to 25.0 cm^3 of an aqueous solution of the weak acid HX

Figure 4 shows how the pH changes during part of this titration at 298 K



0	5	.	5
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Use **Figure 4** to determine the concentration, in mol dm^{-3} , of the weak acid, HX**[3 marks]**Concentration _____ mol dm^{-3}

0	5	.	6
---	---	---	---

Use **Figure 4** to determine the pH of the mixture at half equivalence (half neutralisation).Use this pH to determine the value of the acid dissociation constant, K_a , at 298 K for the weak acid HXGive the units for K_a **[3 marks]**

pH at half equivalence _____

Value of K_a _____Units of K_a _____

12

Turn over for the next question**Turn over ►**

0	6
---	---

This question is about metal ions and complexes.

0	6	.	1
---	---	---	---

Give the full electron configuration of Fe^{2+}

[1 mark]

0	6	.	2
---	---	---	---

In the haemoglobin complex, an Fe^{2+} ion has a co-ordination number of six.

A multidentate haem ligand and an oxygen ligand form co-ordinate bonds to Fe^{2+} enabling oxygen to be transported in the blood.

Define multidentate ligand.

[1 mark]

0	6	.	3
---	---	---	---

The concentration of an Fe^{2+} complex in blood can be measured by colorimetry using light of wavelength 830 nm

Calculate the energy change (ΔE), in J, when a d electron is excited in this Fe^{2+} complex.

Speed of light, $c = 3.00 \times 10^8 \text{ m s}^{-1}$

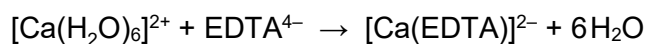
Planck constant, $h = 6.63 \times 10^{-34} \text{ J s}$

[2 marks]

ΔE _____ J



Adding an aqueous solution of EDTA^{4-} to blood removes $\text{Ca}^{2+}(\text{aq})$



0 6 . 4

Explain, in terms of entropy and Gibbs free-energy, why the formation of the metal–EDTA complex in this reaction is **not** reversible.

[3 marks]

0 6 . 5

Describe a test-tube reaction to show that EDTA^{4-} has removed calcium ions from an aqueous sample of calcium chloride with a concentration of $0.100 \text{ mol dm}^{-3}$

State the result of the test.

[2 marks]

Test

Result

9

Turn over ►



0	7
---	---

This question is about transition metals.

0	7	.	1
---	---	---	---

State why the d block metal zinc is **not** a transition metal.

[1 mark]

In the Contact process, vanadium(V) oxide (V_2O_5) acts as a heterogeneous catalyst in the oxidation of sulfur dioxide to sulfur trioxide.

0	7	.	2
---	---	---	---

Write **two** equations to show how V_2O_5 acts as a catalyst in the Contact process.

Explain why the catalyst is heterogeneous.

[3 marks]

Equation 1

Equation 2

Explanation

0	7	.	3
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Suggest why it is important that any impurities are removed from the gas before sulfur dioxide is used in the Contact process.

[1 mark]



Table 5 shows the colours of aqueous solutions of vanadium ions.

Table 5

Aqueous vanadium ion	$\text{VO}_2^+(\text{aq})$	$\text{VO}^{2+}(\text{aq})$	$\text{V}^{3+}(\text{aq})$	$\text{V}^{2+}(\text{aq})$
Colour	Yellow	Blue	Green	Purple

Table 6 shows the standard electrode potentials of aqueous vanadium ions.

Table 6

Electrode half-equation	$E^\ominus / \text{V}$
$\text{VO}_2^+(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \text{VO}^{2+}(\text{aq}) + \text{H}_2\text{O}(\text{l})$	+1.00
$\text{VO}^{2+}(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \text{V}^{3+}(\text{aq}) + \text{H}_2\text{O}(\text{l})$	+0.34
$\text{V}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{V}^{2+}(\text{aq})$	-0.26
$\text{V}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{V}(\text{s})$	-1.20

0 7 . 4

An excess of powdered zinc is added to an aqueous solution containing VO_2^+ ions. This causes a series of colour changes.

When zinc stops reacting, the solution is purple.

Use **Table 5** and **Table 6** to suggest a value for the standard electrode potential for the reduction of Zn^{2+}

Write a half-equation, with state symbols, for the reduction of Zn^{2+}

Write an overall equation for the reaction between zinc and VO_2^+ ions to form the purple solution.

[3 marks]

Electrode potential for reduction of Zn^{2+} _____ V

Half equation

Overall equation

Turn over ►



0 7 . 5 Transition metal complexes can have a range of shapes.

Table 7 shows the different shapes shown by transition metal complexes.

Tick (✓) one box in each row of **Table 7** to show the shape of each transition metal complex.

[2 marks]

Table 7

Complex	Linear	Octahedral	Tetrahedral	Square Planar
FeCl_4^-				
$\text{Pt}(\text{NH}_3)_2\text{Cl}_2$				
$[\text{Ag}(\text{NH}_3)_2]^+$				
$[\text{Cu}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_2(\text{H}_2\text{O})_2]^{2+}$				

10



Turn over for the next question

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0 8

This question is about aluminium.

0 8 . 1

Which block of the Periodic Table contains aluminium?

Tick (✓) **one** box.**[1 mark]**

s block

☐

p block

☐

d block

☐

f block

☐**0 8 . 2**

Give an equation for the reaction of aluminium with oxygen.

[1 mark]

0 8 . 3Aluminium hydroxide, $\text{Al}(\text{H}_2\text{O})_3(\text{OH})_3$, reacts with acids and with alkalis.

State the word that describes this property.

Give an equation to show aluminium hydroxide reacting with an acid.

Give an equation to show aluminium hydroxide reacting with an alkali.

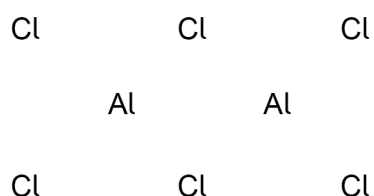
[3 marks]Word to describe the property

Equation for aluminium hydroxide reacting with an acid

Equation for aluminium hydroxide reacting with an alkali



0 8 . 4

Figure 5 shows the atoms in a molecule of Al_2Cl_6 Complete **Figure 5**.Use — to show covalent bonds and $\rightarrow$ to show co-ordinate bonds.**[1 mark]****Figure 5**

0 8 . 5

Explain why $[\text{Al}(\text{H}_2\text{O})_6]^{3+}(\text{aq})$ forms an acidic solution in water.
Include an equation in your answer.

[3 marks]

0 8 . 6

Aqueous sodium carbonate is added to aqueous aluminium sulfate in a test tube.

Describe what is observed in this reaction.

Write an equation.

[3 marks]

Observation(s)

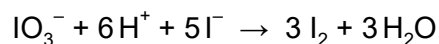
Equation



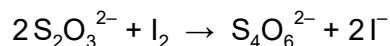
0 9

This question is about a redox titration.

Iodate ions (IO_3^-) react with iodide ions (I^-) in acidic conditions to produce iodine (I_2).



Sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3$) reacts with iodine.



- 0.545 g of KIO_3 is dissolved in water and made up to a 250 cm^3 solution in a volumetric flask.
- A 25.0 cm^3 portion of the solution is transferred to a conical flask.
- An excess of sulfuric acid is added to the conical flask.
- An excess of potassium iodide solution is added to the conical flask.
The solution turns dark brown because iodine is produced.
- Sodium thiosulfate is added from a burette until the solution turns colourless.
This shows that all the iodine has reacted.

Table 8 shows the titration results.

Table 8

	1	2	3
Final volume / cm^3	27.00	47.85	33.10
Initial volume / cm^3	3.65	25.35	10.55
Titre / cm^3	23.35	22.50	22.55

0 9 . 1

Use suitable values from **Table 8** to calculate the mean titre.

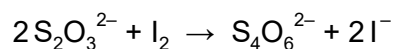
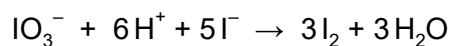
[1 mark]

Mean titre _____ cm^3



09.2

Use these equations to deduce how many moles of $\text{S}_2\text{O}_3^{2-}$ react with the iodine produced from the reaction of one mole of IO_3^-

**[1 mark]**

Moles of $\text{S}_2\text{O}_3^{2-}$ required _____

09.3

Use your answers to Question **09.1** and Question **09.2** to calculate the concentration, in mol dm^{-3} , of the sodium thiosulfate solution.

(If you were unable to answer Question **09.2** then you should assume that the moles of $\text{S}_2\text{O}_3^{2-}$ required is 2.5

This is **not** the correct answer.)

[4 marks]

Concentration _____ mol dm^{-3}

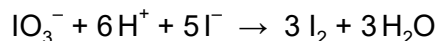
Question 9 continues on the next page

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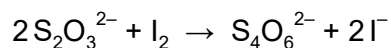


The method from page 22 is repeated below.

Iodate ions (IO_3^-) react with iodide ions (I^-) in acidic conditions to produce iodine (I_2).



Sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3$) reacts with iodine.



- 0.545 g of KIO_3 is dissolved in water and made up to a 250 cm^3 solution in a volumetric flask.
- A 25.0 cm^3 portion of the solution is transferred to a conical flask.
- An excess of sulfuric acid is added to the conical flask.
- An excess of potassium iodide solution is added to the conical flask.
The solution turns dark brown because iodine is produced.
- Sodium thiosulfate is added from a burette until the solution turns colourless.
This shows that all the iodine has reacted.

0 9 . 4

Explain why there should be an excess of potassium iodide and sulfuric acid in the conical flask.

[1 mark]

0 9 . 5

Explain why the exact amount of potassium iodide and sulfuric acid used in the conical flask does **not** need to be known.

[1 mark]

8

END OF QUESTIONS



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Question number	Additional page, if required. Write the question numbers in the left-hand margin.



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